

Beyond Tabular Q-Learning for SAM Prompt Optimization: Deep RL Algorithms, Domain Transfer, and Temporal Prompting

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I. INTRODUCTION

The Segment Anything Model (SAM) [1] introduced a paradigm shift in image segmentation by enabling prompt-driven, class-agnostic mask prediction across virtually any visual domain. Given a set of input prompts such as points, bounding boxes, or text, SAM produces high-quality segmentation masks without requiring task-specific fine-tuning. However, SAM’s performance is highly sensitive to the quality and placement of these prompts. A poorly placed point prompt can yield a mask that captures the wrong region entirely, while a well-placed one produces near-perfect segmentation. This sensitivity creates a critical bottleneck: in automated pipelines where no human operator is available to provide prompts interactively, the segmentation quality depends entirely on the prompt selection strategy.

Liu et al. [2] recently addressed this bottleneck by formulating prompt optimization as a reinforcement learning problem. Their Plug-and-Play Point Prompt Optimizer (PPO) uses tabular Q-learning with epsilon-greedy exploration to iteratively refine point prompt distributions over a dual-space heterogeneous graph constructed from DINOv2 [3] patch features. The graph encodes both feature-space similarity and physical-space proximity between candidate prompt locations, and the agent learns to add or remove prompt nodes to optimize a graph-based proxy reward. The method achieves state-of-the-art results on several benchmarks, surpassing both manual prompt strategies and prior automated approaches. Despite these advances, several significant extensions remain unexplored. The original work uses tabular Q-learning with discrete graph-manipulation actions, leaving open the question of whether deep RL methods with continuous action spaces and neural network policies could improve both generalization and sample efficiency. The proxy reward based on graph distances does not directly optimize segmentation quality, suggesting that direct IoU-based rewards could better align training with the downstream objective. The evaluation is restricted to standard computer vision benchmarks, leaving domain transfer to specialized fields such as medical imaging and remote sensing untested. The method operates on static images, with no adaptation for temporally consistent video segmentation via SAM 2 [4].

We propose to replicate the Plug-and-Play PPO framework and extend it along four axes. First, we replace the tabular Q-learning agent with deep RL algorithms—Proximal Policy Optimization (PPO) [5], Soft Actor-Critic (SAC) [6], and Twin Delayed DDPG (TD3) [7]—using neural network policies that predict continuous point coordinates directly, enabling generalization across images. Second, we evaluate domain transfer to medical polyp segmentation (KVASIR-SEG [8]) and remote sensing imagery [9]. Third, we adapt the prompt optimizer for SAM 2 video segmentation with temporal consistency constraints. Fourth, we replace the graph-distance proxy reward with direct IoU-based feedback augmented by boundary-weighted reward shaping [10] to improve mask quality on thin and irregular structures.

II. APPROACH

A. Background: Plug-and-Play PPO

The original Plug-and-Play Point Prompt Optimizer (PPO) [2] reformulates prompt optimization as a heterogeneous graph optimization problem solved via reinforcement learning. Given a reference image with a known mask and a target image, DINOv2 ViT-G/14 [3] extracts patch-level features (one 1024-dimensional vector per 14×14 patch on 560×560 images, yielding $40 \times 40 = 1600$ patches). Feature matching via between reference foreground/background patches and target patches generates initial positive and negative prompt candidates. The system then constructs a dual-space heterogeneous multigraph G where nodes represent prompt points and five edge types encode relationships across two spaces: *feature-space* edges capture DINOv2 cosine distances between positive-positive ($e_{\text{feat,pos}}$), positive-negative ($e_{\text{feat,cross}}$) node pairs, while *physical-space* edges capture Euclidean pixel distances between positive-positive ($e_{\text{phys,pos}}$), negative-negative ($e_{\text{phys,neg}}$), and cross-label ($e_{\text{phys,cross}}$) pairs.

The RL agent operates over this graph via tabular Q-learning with epsilon-greedy exploration (initial $\epsilon = 1.0$, decay factor 0.995, minimum 0.1) and experience replay (buffer size 10,000, batch size 64). At each step t , the state s_t is the current graph G_t . Actions are discrete graph manipulations: *remove_pos*, *remove_neg*, *restore_pos*, *restore_neg*, or *add* a new node. Node counts are constrained between 5 and 20 per category. The reward is a *graph-distance proxy* rather than direct

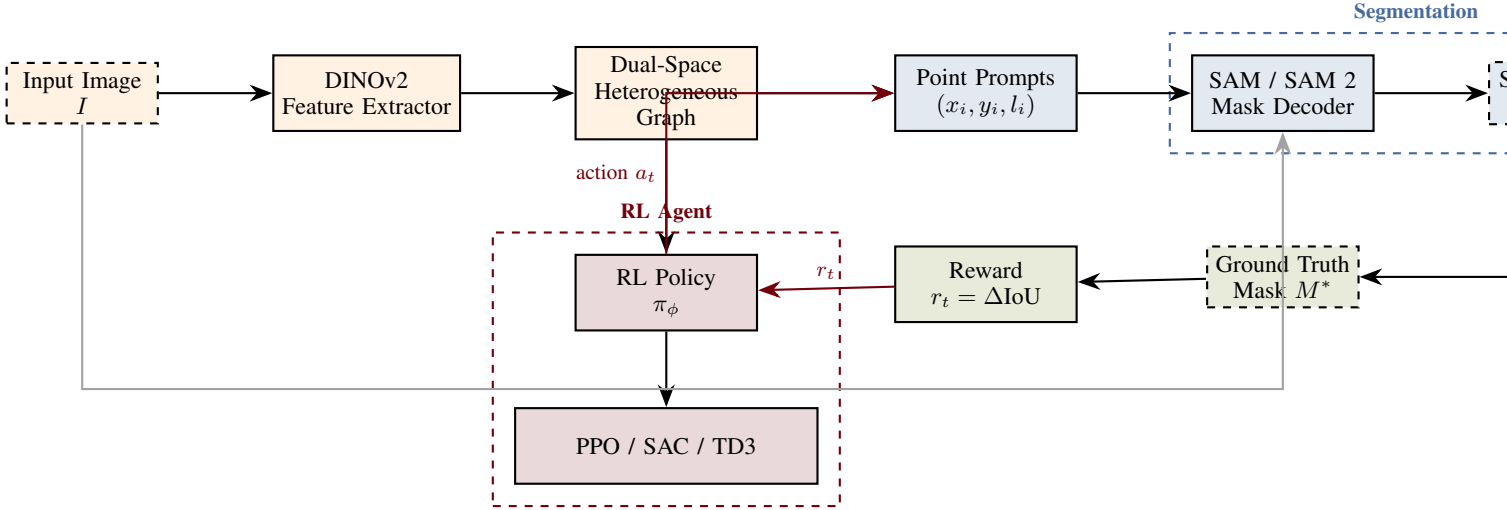


Fig. 1: System architecture of the extended Plug-and-Play prompt optimization framework. An input image is processed by DINOv2 to construct a dual-space heterogeneous graph encoding feature similarity and spatial proximity. The RL policy (PPO, SAC, or TD3) selects point prompt coordinates and labels, which are passed to SAM or SAM 2 for mask generation. The segmentation quality improvement (IoU delta) serves as the reward signal, closing the optimization loop.

segmentation quality. Feature-space terms reward decreased positive-positive distances (semantic coherence) and increased positive-negative distances (discriminability), weighted $5\times$ more heavily than physical-space terms that reward spatial spread among same-label nodes and tight boundaries between opposite-label nodes. The *add* action incurs a penalty of -10 to discourage unnecessary expansion. Training runs for 100,000 episodes with a maximum of 100 steps per episode, learning rate $\alpha = 0.1$, and discount factor $\gamma_{rl} = 0.9$.

B. Extension 1: Deep RL Algorithm Comparison

The original PPO uses tabular Q-learning, which stores Q-values for each observed state-action pair without function approximation. This approach cannot generalize across images: each new image requires loading a separate Q-table, and the discrete action space (five graph operations) limits the precision of prompt placement. We propose replacing the tabular agent with three deep RL algorithms that use neural network policies capable of generalizing across images and predicting continuous point coordinates directly.

Proximal Policy Optimization (PPO) [5] is an on-policy algorithm that constrains policy updates via a clipped surrogate objective, preventing destructively large updates. We use PPO as the deep RL baseline with a shared encoder that processes the dual-space graph features through a graph neural network, outputting a stochastic policy over continuous actions.

Soft Actor-Critic (SAC) [6] maximizes a maximum-entropy objective $J(\pi) = \sum_t \mathbb{E}_{(s_t, a_t) \sim \rho_\pi} [r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot|s_t))]$, where α controls the entropy bonus $\mathcal{H}$. The entropy term encourages exploration of diverse prompt placements, which is particularly valuable in early training when the policy has not yet learned which regions are informative. SAC uses two Q-networks to mitigate overestimation bias and updates the

policy via the reparameterization trick for continuous actions. As an off-policy method, SAC reuses past transitions via a replay buffer, offering substantially better sample efficiency than on-policy PPO.

Twin Delayed DDPG (TD3) [7] extends DDPG with three modifications: clipped double Q-learning to reduce value over-estimation, delayed policy updates to reduce per-update error, and target policy smoothing to prevent the policy from exploiting narrow peaks in the Q-function. TD3 uses a deterministic policy with additive exploration noise, contrasting with SAC’s stochastic policy.

For all three deep RL algorithms, we parameterize the action space as $a_t = (x, y, p) \in [0, 1]^3$, where (x, y) are normalized image coordinates and p is thresholded to determine the prompt label ($l = +1$ if $p > 0.5$, else $l = -1$). This continuous formulation allows sub-patch precision in prompt placement, unlike the original discrete grid-aligned actions. We additionally retain the original tabular Q-learning agent from [2] as a direct replication baseline.

C. Extension 2: Domain Transfer

A key claim of SAM is zero-shot generalizability across visual domains. However, prompt sensitivity varies dramatically across domains. Medical images exhibit low contrast and ambiguous boundaries [11], while remote sensing imagery contains densely packed objects at multiple scales [9]. We evaluate whether a prompt optimizer trained on natural images (Pascal VOC [12]) transfers to these specialized domains, and whether domain-specific fine-tuning of the RL policy improves results. AlignSAM [13] demonstrated that RL can align SAM to open-context scenarios via reinforcement learning from human feedback, motivating our hypothesis that RL-based prompt optimization can similarly adapt across domains.

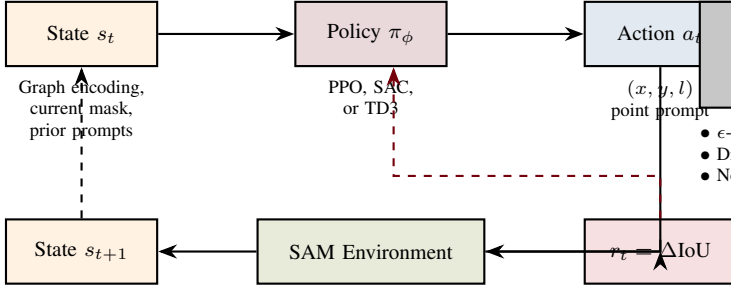


Fig. 2: MDP formulation for RL-based prompt optimization. The state encodes the dual-space graph, current mask, and prior prompts. The policy selects point coordinates and labels. SAM produces a new mask, and the IoU improvement serves as the reward.

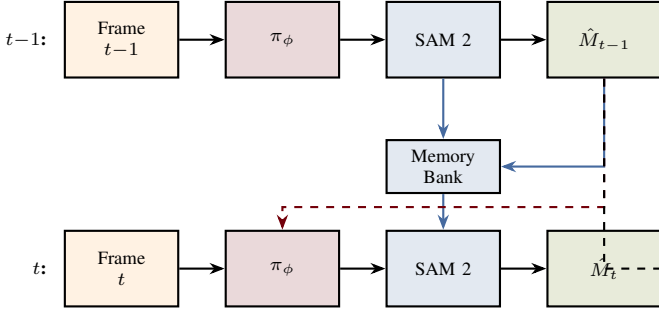


Fig. 3: Temporal prompt optimization with SAM 2. The RL policy receives temporal context from previous frames via the memory bank and prior mask history. The reward includes a temporal consistency (TC) bonus to encourage smooth mask predictions across frames.

D. Extension 3: Temporal Prompting with SAM 2

SAM 2 [4] extends the original SAM architecture to video with a memory-augmented streaming design. A memory bank stores spatial features and object pointers from previous frames, and a memory attention module conditions the current frame’s features on this temporal context. SAM 2 achieves state-of-the-art video object segmentation while running at approximately 44 frames per second. Recent work on SAM2RL [14] demonstrates that RL can optimize SAM 2’s memory bank management, achieving improvements exceeding three times the gains of hand-crafted heuristics.

We propose adapting the prompt optimizer for video input by augmenting the state representation with temporal features. At frame t , the state includes not only the current frame’s dual-space graph but also a summary of the previous frame’s mask and prompt history. The reward is modified to include a temporal consistency term $r_t = \Delta \text{IoU}_t + \beta \cdot \text{TC}_t$, where TC_t measures the smoothness of mask boundaries across consecutive frames. This encourages the policy to select prompts that maintain stable object tracking rather than optimizing each frame independently.

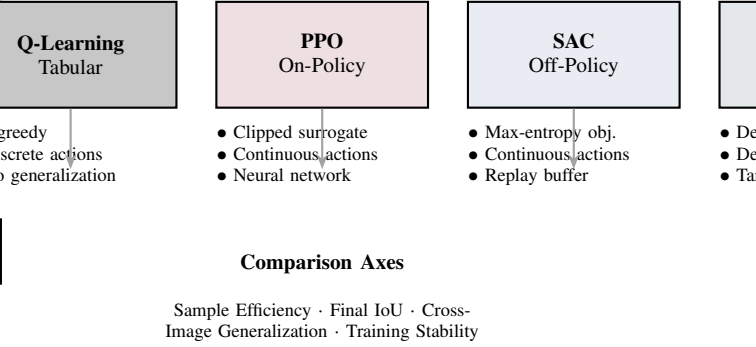


Fig. 4: Comparison of four RL approaches for prompt optimization. The original tabular Q-learning [2] uses discrete graph operations without cross-image generalization. Our proposed deep RL alternatives (PPO, SAC, TD3) use neural network policies with continuous action spaces for direct coordinate prediction.

E. Extension 4: Direct IoU Reward with Boundary Shaping

The original PPO uses a graph-distance proxy reward that optimizes feature-space and physical-space relationships between prompt nodes rather than directly measuring segmentation quality. While this proxy correlates with good prompt placement, it introduces an alignment gap between what the agent optimizes and what ultimately matters: mask accuracy. Our first modification replaces this proxy with a *direct* IoU reward, $r_t = \text{IoU}(\hat{M}_t, M^*) - \text{IoU}(\hat{M}_{t-1}, M^*)$, computed by running SAM after each prompt action and comparing the resulting mask against ground truth.

Building on this direct reward, we further introduce boundary-weighted shaping following principles from [10]. Standard IoU treats all pixels equally, which can underweight boundary accuracy for objects with thin protrusions or irregular edges. Let B^* denote the set of boundary pixels in the ground truth mask (obtained via morphological gradient) and $\hat{B}$ the corresponding boundary in the predicted mask. The shaped reward is defined as

$$r_t^{\text{shaped}} = \Delta \text{IoU}_t + \gamma \cdot \Delta \text{BIOU}_t \quad (1)$$

where BIOU computes IoU restricted to a narrow band around object boundaries and γ controls the boundary emphasis. For class-imbalanced datasets such as KVASIR-SEG where polyps vary dramatically in size, we additionally weight the reward inversely proportional to object area, preventing the policy from ignoring small objects.

III. EVALUATION

A. Datasets

We evaluate on four datasets spanning diverse visual domains. Pascal VOC [12] provides 1,464 training and 1,449 validation images with 20 object categories, serving as the primary natural-image benchmark. FSS-1000 [15] contains 1,000 classes with 10 images per class for few-shot segmentation evaluation. KVASIR-SEG [8] provides 1,000 gastrointestinal

TABLE I: Evaluation plan across datasets and proposed extensions.

Dataset	Algo.	Domain	Video	Reward
Pascal VOC	✓	—	—	✓
FSS-1000	✓	—	—	✓
KVASIR-SEG	✓	✓	—	✓
DAVIS 2017	✓	—	✓	✓

polyp images for medical domain transfer evaluation. For video segmentation, we use the DAVIS 2017 benchmark, which contains 90 video sequences with dense per-frame annotations.

B. Baselines

We compare our extended methods against seven baselines spanning non-RL and RL approaches. *Center-of-mass prompts* place a single positive point at the centroid of a coarse initial mask. *Random point prompts* sample points uniformly from the image. *Grid-based prompts* place points on a regular grid with labels assigned by a coarse mask. *Greedy heuristic* selects the point that maximizes instantaneous IoU at each step via exhaustive search over a candidate set. *Original tabular Q-learning* [2] provides the direct replication baseline with the graph-distance proxy reward and discrete actions. *FocalClick* [16] represents the supervised interactive segmentation baseline, and *RITM* [17] provides an iterative mask-guided training baseline.

C. Metrics

The primary segmentation quality metrics are Intersection-over-Union (IoU) and Dice coefficient, reported as mean values across test images. We additionally report Boundary IoU (BIOU) to assess edge quality, the number of prompt iterations required to reach a target IoU threshold (efficiency), and wall-clock inference time per image. For the algorithm comparison, we measure sample efficiency as the number of environment interactions required to reach 90% of final performance, and training stability as the variance of IoU across five random seeds. For video segmentation, we report the $\mathcal{J}\&\mathcal{F}$ metric (mean of region similarity $\mathcal{J}$ and contour accuracy $\mathcal{F}$) following standard VOS evaluation protocol.

D. Ablation Studies

We conduct the following ablations to isolate contributions of each component. Removing the dual-space graph and using raw image features as state measures the value of the heterogeneous graph encoding. Removing the boundary-weighted reward term measures the effect of reward shaping on boundary quality. For the video extension, removing the temporal consistency bonus measures whether temporal awareness improves tracking stability. Varying the entropy coefficient α in SAC and the exploration noise scale in TD3 quantifies the sensitivity of each algorithm to exploration hyperparameters.

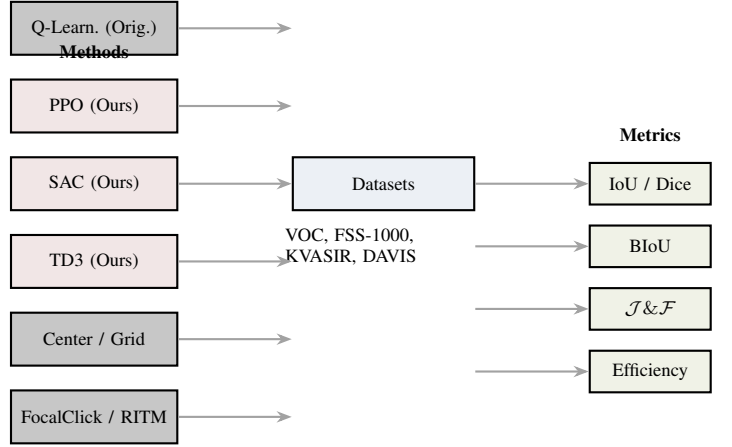


Fig. 5: Evaluation pipeline. RL-based methods (PPO, SAC, TD3) and non-RL baselines are evaluated across four datasets using segmentation quality, boundary accuracy, video tracking, and prompt efficiency metrics.

TABLE II: Algorithmic properties of each RL method.

Property	Q-Learn.	PPO	SAC	TD3
Policy Type	ϵ -greedy	Stochastic	Stochastic	Determ.
Function Approx.	Tabular	Neural net	Neural net	Neural net
On/Off-Policy	Off-policy	On-policy	Off-policy	Off-policy
Action Space	Discrete	Continuous	Continuous	Continuous
Replay Buffer	Yes (10K)	No	Yes	Yes
Entropy Bonus	No	No	Yes	No
Generalizes	No	Yes	Yes	Yes

IV. CONCLUSION

We propose to replicate and substantially extend the Plug-and-Play PPO framework for RL-based point prompt optimization in SAM segmentation. Our extensions address four fundamental limitations of the original tabular Q-learning approach: replacing it with deep RL algorithms (PPO, SAC, TD3) that use neural network policies with continuous action spaces for cross-image generalization, evaluating domain transfer to medical and remote sensing imagery, adapting the prompt optimizer for temporally consistent video segmentation with SAM 2, and replacing the graph-distance proxy reward with direct IoU feedback augmented by boundary-weighted shaping. The evaluation plan covers four datasets (Pascal VOC, FSS-1000, KVASIR-SEG, DAVIS 2017), seven baselines including the original tabular agent, and five metric families, with ablation studies to isolate each component's contribution. This work will provide empirical evidence for whether deep RL with continuous actions can outperform tabular Q-learning for prompt optimization, whether learned prompt policies generalize across visual domains, and whether direct segmentation rewards with temporal awareness yield measurable improvements. The project leverages an existing published codebase [2] with standard benchmarks, providing a low-risk foundation for these novel extensions.

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